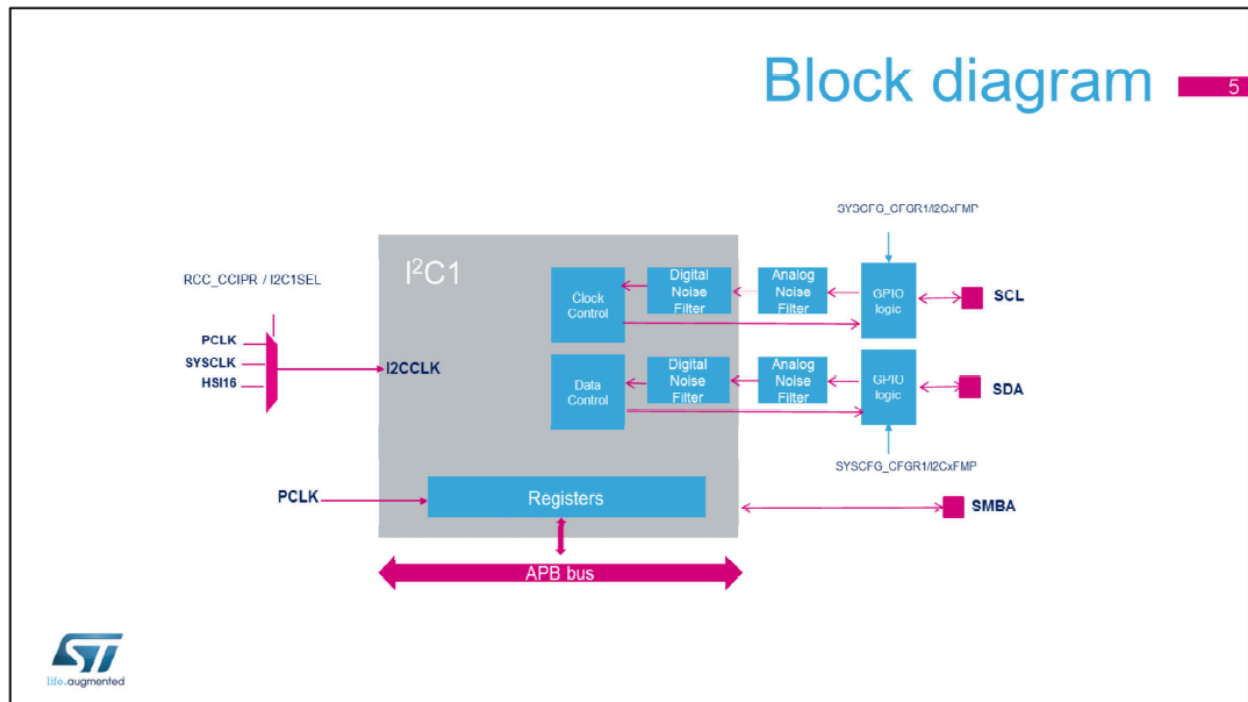


I²C (Inter Integrated Circuits):



I²C (pronounced "I-squared-C") is a **very simple** and **popular communication protocol** used to connect microcontrollers with:

- Sensors
- Memories
- Other microcontrollers
-

Only **two wires** are needed:

Line	Purpose
SCL	Serial Clock Line (carries the clock)
SDA	Serial Data Line (carries the data)

 **Major Point: Only 2 wires** for communication between multiple devices!

Here also there can be multiple slaves but there is no chip select(or SS) pin. Here address bits are used to differentiate. {7 bit to 10 bit addressing modes}.

Here there also can be multiple masters in I2C which was not there in SPI.

Push-Pull GPIO in SPI

Open drain in I2C

Then why we still need resistor in half duplex?

In half-duplex SPI, both the master and slave can drive the **same data line** (e.g., **MISO/MOSI shared**). If one device drives the line **high** while the other drives it **low** (due to misconfiguration or timing issues), that creates a **short circuit** — high current flows between the devices.

A **small series resistor** (e.g., 100–330 ohms) helps **limit that current**, protecting the GPIO pins from damage.

Even with push-pull drivers, a resistor in half-duplex SPI is a **safety and robustness** measure to:

- Prevent damage from **bus contention**
- Improve **signal quality**
- Protect against **ESD/overvoltage**

Features of STM32G0 I²C

- Follows standard **I²C**, **SMBus**, and **PMBus** specifications
- Supports:
 - **Master** and **Slave** modes
 - **Multi-master** (more than one master)
 - **7-bit and 10-bit addressing**

- Can wake MCU up from **Stop Mode** when needed
- Works even when system clock is stopped (using independent clock)

Basic I²C Operation

♦ Open-Drain Pins

- I²C uses **open-drain outputs**: devices can only **pull the line low**.
- **External pull-up resistors** are needed to bring lines HIGH.

 **Major Point:** Without pull-up resistors, **communication can't happen**.

How I²C Communication Works

1. **Start Condition:** Master pulls SDA low while SCL is high.
2. **Addressing:** Master sends the 7-bit (or 10-bit) address of the slave.
3. **Acknowledgment:** Slave acknowledges by pulling SDA low.
4. **Data Transfer:** Bytes are transferred (Master to Slave or Slave to Master).
5. **Stop Condition:** Master releases SDA while SCL is high.

 **Major Point:** Start → Address → Data → Stop

Important Configurations

♦ Timing Control

- **Setup and Hold Times** can be programmed.
- Ensures correct time gaps between clock and data.

◆ Noise Filters

- Analog and Digital filters remove glitches from SCL and SDA lines.

◆ Clock Stretching

- Slaves can **hold the clock low** if they need more time to process data.

🧩 Addressing

- Supports **7-bit** and **10-bit** addresses.
- Slave can have **multiple addresses** using masking options.

🌙 Low Power Wakeup Feature

- MCU can **sleep** (Stop Mode) to save power.
- If **address match** happens, I²C can **wake up** the MCU automatically.

🟡 **Major Point:** I²C helps to save energy by waking the chip only when needed.

Applications:

SMBUS: System Management BUS (ex- Battery Management System)
(older version) (tighter timing and voltage specs)

PMBUS: Power Management Bus (supports both linear and direct data formats)

I²C-> Running on PCLK

PCLK/2 is max speed for SPI to communicate (in MHz).

For I²C, Operations upto 400K Hz are there and 1M hz coming.

SPI vs I2C

 Feature	SPI	I ² C	
1. Wires Required	4 (MOSI, MISO, SCK, SS)	2 (SDA, SCL)	
2. Speed	Faster (up to tens of MHz)	Slower (usually up to 1 MHz)	
3. Communication	Full-Duplex	Half-Duplex	
4. Master/Slave	Single master, multiple slaves	Multi-master, multi-slave	
5. Addressing	No addressing, uses SS lines	7-bit or 10-bit hardware addressing	
6. Wiring Complexity	More wires, harder to expand	Fewer wires, easier to expand	
7. Bus Arbitration	Not supported	Supported (handles collisions)	
8. Use Case	High-speed devices (e.g. displays)	Low-speed sensors (e.g. RTC, temp)	

UART vs I2C

 Feature	UART (Universal Asynchronous Receiver/Transmitter)	I ² C (Inter-Integrated Circuit)
1. Wires Required	2 (TX, RX)	2 (SDA, SCL)
2. Speed	Moderate (commonly 9600–115200 bps, up to Mbps)	Slower (typically ≤ 1 MHz)
3. Communication	Point-to-point (1-to-1)	Multi-master, multi-slave
4. Clock Required	✗ No clock (asynchronous)	✓ Yes, uses a shared clock (SCL)
5. Addressing	✗ No addressing, just direct connection	✓ Yes, uses 7/10-bit addressing
6. Hardware Simplicity	Very simple hardware	Slightly more complex hardware
7. Data Checking	Parity (optional)	PEC/CRC (optional with SMBus)
8. Use Case	Simple 1-to-1 serial communication (e.g. PC ↔ MCU)	Talking to multiple devices on shared bus